

Sri Lanka Institute of Information Technology



Lab Submission- 02

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Web based Market Store

Zentra Mart

Software Engineering | SE2030

B.Sc. (Hons) in Information Technology

Project Overview

Zentra Mart is a web-based marketplace platform that connects product listing, buyer search, shopping cart and checkout, order tracking, and platform administration. It connects buyers and sellers on one shared platform, allowing sellers to list and manage products while buyers search, purchase, and track their orders. The system is intended to reduce the fragmented, manual tools (social media posts, chat apps, spreadsheets) that individual sellers currently rely on, and to give both buyers and sellers a single, trustworthy place to transact.

This report uses the same Zentra Mart project topic, requirements context, CRUD ownership, and stakeholder personas identified at the requirements gathering phase. The priorities, role allocation, and sprint sequence below are the planning baseline.

Core Scope Used for Sprint Planning

- Account registration, role selection (buyer/seller), profile management and password recovery.
- Product listing creation, image upload, pricing, stock management and listing lifecycle (active/sold/archived).
- Product search, category and price filtering, sorting and product detail viewing.
- Shopping cart management, order summary, and simulated checkout/payment processing.
- Order status tracking, order history, and seller-side order fulfilment updates.
- Role-based admin oversight, account/listing moderation, audit visibility and authorized platform reporting.

Group Members and Scrum Roles

Member	Student ID	Scrum Role	Core Ownership / Function
Jayawardena S.D.D.L.	IT25101719	Product Owner	User Registration & Account Management
Madugalle E.W.M.W.W.S.A.K.	IT25101711	Scrum Master	Product Listing Management
Alahakoon A.R.S.S.	IT25101690	Developer	Search & Browse / Product Catalog
Sooriyaarachchi A.G.S.T.	IT25101736	Developer	Shopping Cart & Checkout
Jayasinghe J.M.N.R.N.	IT25101657	Developer	Order Management & Tracking
Ahamed M. R. S.	IT25101678	Developer	Admin Management Panel

Scrum roles note

Product Owner: owns value ordering, clarifies acceptance criteria and accepts finished work.

Scrum Master: facilitates Scrum events, makes blockers visible and helps the team follow the agreed process.

Developers: build, test, integrate and demonstrate selected sprint work as a cross-functional team.

Ownership rule: Scrum accountability does not replace the CRUD division at the requirement gathering phase. Jayawardena S.D.D.L. retains User Registration & Account Management and Madugalle E.W.M.W.S.A.K. retains Product Listing Management. A functional owner defines domain rules and verifies behavior; Developers implement and test; only the Product Owner accepts a completed backlog item.

Identified Personas

No.	Persona	Primary Need	Owner
1	Registered User (Buyer / Seller)	Creates an account, selects a role, and manages profile details securely.	IT25101719
2	Seller (Vendor)	Lists and manages products for sale on the platform.	IT25101711
3	Buyer (Shopper)	Searches, filters and browses products to find what they need.	IT25101690
4	Buyer (Checkout Customer)	Adds items to a cart and completes a purchase smoothly.	IT25101736
5	Order Management User (Buyer & Seller)	Sellers fulfil orders; buyers track order status and history.	IT25101657
6	Platform Administrator	Oversees accounts, moderates listings and reviews platform reports.	IT25101678

User Stories by Persona

Each story expresses one independently testable business outcome. Supporting rules and edge cases are handled in acceptance criteria and sprint tasks instead of being bundled into the story statement.

Persona 1: Registered User- IT25101719

Primary need: Creates an account, selects a role, and manages profile details securely.

ID	User Story	Priority
PBI-01	As a registered user, I want to create an account with my email and password so that I can access the platform.	High
PBI-02	As a registered user, I want to choose my role as buyer or seller during sign-up so that I get the correct dashboard and permissions.	High
PBI-03	As a registered user, I want to reset my forgotten password so that I can regain access to my account.	Medium
PBI-04	As a registered user, I want to edit my profile details so that my account information stays accurate.	Low

Persona 2: Seller (Vendor)- IT25101711

Primary need: Lists and manages products for sale on the platform.

ID	User Story	Priority
PBI-05	As a seller, I want to create a new product listing with images, price and stock so that buyers can see and purchase my product.	High
PBI-06	As a seller, I want to edit an existing listing's details so that my product information stays accurate.	Medium
PBI-07	As a seller, I want to mark an item as sold or unavailable so that buyers do not purchase out-of-stock products.	Medium
PBI-08	As a seller, I want to delete a listing I no longer sell so that my catalogue stays current.	Low

Persona 3: Buyer (Shopper)- IT25101690

Primary need: Searches, filters and browses products to find what they need.

ID	User Story	Priority
PBI-09	As a buyer, I want to search products by keyword so that I can quickly find what I am looking for.	High
PBI-10	As a buyer, I want to filter search results by category and price range so that I can narrow down relevant options.	Medium
PBI-11	As a buyer, I want to sort products by price, newest, or rating so that I can compare choices easily.	Medium
PBI-12	As a buyer, I want to view a product's detail page so that I can review its full description before buying.	High

Persona 4: Buyer (Checkout Customer)-IT25101736

Primary need: Adds items to a cart and completes a purchase smoothly.

ID	User Story	Priority
PBI-13	As a buyer, I want to add a product to my cart so that I can purchase it later.	High
PBI-14	As a buyer, I want to adjust the quantity of an item in my cart so that I order the correct amount.	Medium
PBI-15	As a buyer, I want to view an order summary before paying so that I can confirm my purchase details.	Medium
PBI-16	As a buyer, I want to complete checkout with a simulated payment so that my order is placed successfully.	High

Persona 5: Order Management User (Buyer & Seller)- IT25101657

Primary need: Sellers fulfil orders; buyers track order status and history.

ID	User Story	Priority
PBI-17	As a seller, I want to view incoming orders for my products so that I can fulfil them promptly.	High
PBI-18	As a seller, I want to update an order's status (pending → shipped → delivered) so that buyers stay informed.	High
PBI-19	As a buyer, I want to view my order history so that I can track my past purchases.	Medium
PBI-20	As a buyer, I want to check the current status of an active order so that I know when to expect delivery.	Low

Persona 6: Platform Administrator- IT25101678

Primary need: Oversees accounts, moderates listings and reviews platform reports.

ID	User Story	Priority
PBI-21	As an administrator, I want to view and manage user accounts so that I can suspend accounts that violate policy.	High
PBI-22	As an administrator, I want to moderate product listings so that inappropriate or fraudulent listings are removed.	Medium
PBI-23	As an administrator, I want to view platform-wide reports (users, listings, orders) so that I can monitor marketplace health.	Medium
PBI-24	As an administrator, I want to review escalated complaints so that disputes between buyers and sellers are resolved fairly.	Low

Product Backlog

The backlog is ordered first by business priority and then by dependency. High denotes core account, listing, search, checkout, or order-fulfilment capability; Medium denotes essential supporting capability; and Low denotes recovery or reporting enhancements that follow the operational core.

Priority Rationale and Distribution

Priority	Count	Selection Rule	Planned Position
High	10	Core account, listing, search, checkout, or order-fulfilment capability	Earliest dependency-valid sprint
Medium	10	Essential dependent operational capability	Sprints 1–3 by dependency
Low	4	Deferrable recovery or reporting enhancement	Sprint 4

Ranked Product Backlog

Rank	ID	Persona	User Story (short)	Priority
1	PBI-01	Registered User	Create an account with email and password	High
2	PBI-02	Registered User	Choose role as buyer or seller at sign-up	High
3	PBI-21	Platform Administrator	View and manage user accounts	High
4	PBI-05	Seller	Create a new product listing	High
5	PBI-09	Buyer	Search products by keyword	High
6	PBI-12	Buyer	View a product's detail page	High
7	PBI-13	Buyer	Add a product to cart	High
8	PBI-16	Buyer	Complete checkout with simulated payment	High
9	PBI-06	Seller	Edit an existing listing's details	Medium
10	PBI-07	Seller	Mark an item as sold or unavailable	Medium

11	PBI-10	Buyer	Filter search results by category and price	Medium
12	PBI-15	Buyer	View order summary before paying	Medium
13	PBI-17	Seller	View incoming orders	High
14	PBI-18	Seller	Update an order's status	High
15	PBI-19	Buyer	View order history	Medium
16	PBI-22	Platform Administrator	Moderate product listings	Medium
17	PBI-23	Platform Administrator	View platform-wide reports	Medium
18	PBI-03	Registered User	Reset a forgotten password	Medium
19	PBI-04	Registered User	Edit profile details	Low
20	PBI-08	Seller	Delete a listing no longer sold	Low
21	PBI-11	Buyer	Sort products by price, newest, or rating	Medium
22	PBI-14	Buyer	Adjust quantity of a cart item	Medium
23	PBI-20	Buyer	Check current status of an active order	Low
24	PBI-24	Platform Administrator	Review escalated complaints	Low

Sprint Planning

The 24 product backlog items are divided across four sprints. Each sprint contains six stories and follows the ranked dependency path from account and listing foundations through search, checkout, order fulfilment, and administrative oversight.

Sprint	Focus	Goal	Stories
Sprint 1	Foundation: accounts, first listing, first search	Establish account creation, role selection, first product listing, and a basic passenger-to-buyer search-to-detail flow.	PBI-01, PBI-02, PBI-21, PBI-05, PBI-09, PBI-12
Sprint 2	Cart, checkout, and listing refinement	Secure the add-to-cart-to-payment flow and complete core listing management and search filtering.	PBI-13, PBI-16, PBI-06, PBI-07, PBI-10, PBI-15
Sprint 3	Order fulfilment, tracking, and moderation	Deliver seller-side order fulfilment, buyer order visibility, and admin moderation and reporting.	PBI-17, PBI-18, PBI-19, PBI-22, PBI-23, PBI-03
Sprint 4	Recovery, enhancement, and release readiness	Complete remaining profile, listing, sorting, cart, tracking, and complaint-handling features; demonstrate release readiness.	PBI-04, PBI-08, PBI-11, PBI-14, PBI-20, PBI-24

1st Sprint

Sprint Goal: Establish account creation, role selection, first product listing, and a basic buyer search-to-detail flow.

Dependency order: an account and role precede any seller listing or buyer action; a published listing precedes search and detail viewing; admin account oversight is established alongside user creation.

Selected User Stories

Sprint ID	PBI	User Story	Priority
SP1-01	PBI-01	As a registered user, I want to create an account with my email and password so that I can access the platform.	High
SP1-02	PBI-02	As a registered user, I want to choose my role as buyer or seller at sign-up so that I get the correct dashboard and permissions.	High
SP1-03	PBI-21	As an administrator, I want to view and manage user accounts so that I can suspend accounts that violate policy.	High
SP1-04	PBI-05	As a seller, I want to create a new product listing with images, price and stock so that buyers can see and purchase my product.	High
SP1-05	PBI-09	As a buyer, I want to search products by keyword so that I can quickly find what I am looking for.	High
SP1-06	PBI-12	As a buyer, I want to view a product's detail page so that I can review its full description before buying.	High

Sprint Backlog — Task Assignments and Ownership

Task ID	Linked PBI	Task	Owner	Hrs
T-01.1	All selected PBIs	Order the selected PBIs, confirm acceptance criteria, review evidence, and accept or reject the increment.	IT25101719 (Product Owner)	4
T-01.2	PBI-01, PBI-02	Build account registration and role-selection with validation and unit tests.	IT25101719 (Account owner)	7
T-01.3	All selected PBIs	Facilitate planning, Daily Scrum, review, retrospective, board updates, and impediment removal.	IT25101711 (Scrum Master)	4
T-01.4	PBI-05	Build listing creation with image upload, price, and stock validation and unit tests.	IT25101711 (Listing owner)	7
T-01.5	PBI-21	Build admin account view/manage screen with suspend/activate logic and access-control tests.	IT25101678 (Admin owner)	7
T-01.6	PBI-09	Build the keyword search query and tests for relevant result matching.	IT25101690 (Search owner)	7
T-01.7	PBI-12	Build the product detail page view and tests for correct data display.	IT25101690 (Search owner)	7
T-01.8	All Sprint 1 PBIs	Execute sprint integration, access-control, regression, and acceptance-evidence checks.	IT25101736 (Developer)	6

2nd Sprint

Sprint Goal: Secure the add-to-cart-to-payment flow and complete core listing management and search filtering.

Dependency order: an existing listing precedes editing/availability changes; cart addition precedes order summary and checkout; filtering builds on the Sprint 1 search foundation.

Selected User Stories

Sprint ID	PBI	User Story	Priority
SP2-01	PBI-13	As a buyer, I want to add a product to my cart so that I can purchase it later.	High
SP2-02	PBI-16	As a buyer, I want to complete checkout with a simulated payment so that my order is placed successfully.	High
SP2-03	PBI-06	As a seller, I want to edit an existing listing's details so that my product information stays accurate.	Medium
SP2-04	PBI-07	As a seller, I want to mark an item as sold or unavailable so that buyers do not purchase out-of-stock products.	Medium
SP2-05	PBI-10	As a buyer, I want to filter search results by category and price range so that I can narrow down relevant options.	Medium
SP2-06	PBI-15	As a buyer, I want to view an order summary before paying so that I can confirm my purchase details.	Medium

Sprint Backlog — Task Assignments and Ownership

Task ID	Linked PBI	Task	Owner	Hrs
T-02.1	All selected PBIs	Order the selected PBIs, confirm acceptance criteria, review evidence, and accept or reject the increment.	IT25101719 (Product Owner)	4
T-02.2	All selected PBIs	Facilitate Scrum events, dependency checks, quality-risk tracking, board updates, and impediment removal.	IT25101711 (Scrum Master)	4
T-02.3	PBI-06	Build listing-edit functionality with validation and unit tests.	IT25101711 (Listing owner)	7
T-02.4	PBI-07	Build sold/unavailable status toggle with tests for buyer-side visibility.	IT25101711 (Listing owner)	7
T-02.5	PBI-10	Build category and price-range filtering with tests for result accuracy.	IT25101690 (Search owner)	7
T-02.6	PBI-13	Build add-to-cart functionality with quantity and stock-check tests.	IT25101736 (Cart owner)	7
T-02.7	PBI-15	Build the order summary view with tests for total-price accuracy.	IT25101736 (Cart owner)	7
T-02.8	PBI-16	Build the simulated checkout/payment flow with success and failure-path tests.	IT25101736 (Cart owner)	7
T-02.9	All Sprint 2 PBIs	Execute cart-to-checkout integration, regression, and acceptance-evidence checks.	IT25101657 (Developer)	6

3rd Sprint

Sprint Goal: Deliver seller-side order fulfilment, buyer order visibility, and admin moderation and reporting.

Dependency order: a placed order (Sprint 2) precedes fulfilment and status updates; status updates precede buyer-side order history; account and listing foundations (Sprint 1) support moderation and reporting.

Selected User Stories

Sprint ID	PBI	User Story	Priority
SP3-01	PBI-17	As a seller, I want to view incoming orders for my products so that I can fulfil them promptly.	High
SP3-02	PBI-18	As a seller, I want to update an order's status (pending → shipped → delivered) so that buyers stay informed.	High
SP3-03	PBI-19	As a buyer, I want to view my order history so that I can track my past purchases.	Medium
SP3-04	PBI-22	As an administrator, I want to moderate product listings so that inappropriate or fraudulent listings are removed.	Medium
SP3-05	PBI-23	As an administrator, I want to view platform-wide reports so that I can monitor marketplace health.	Medium
SP3-06	PBI-03	As a registered user, I want to reset my forgotten password so that I can regain access to my account.	Medium

Sprint Backlog — Task Assignments and Ownership

Task ID	Linked PBI	Task	Owner	Hrs
T-03.1	All selected PBIs	Order the selected PBIs, confirm acceptance criteria, review evidence, and accept or reject the increment.	IT25101719 (Product Owner)	4
T-03.2	PBI-03	Build password-reset flow with token validation and unit tests.	IT25101719 (Account owner)	7
T-03.3	All selected PBIs	Facilitate Scrum events, quality reviews, risk tracking, board updates, and impediment removal.	IT25101711 (Scrum Master)	4
T-03.4	PBI-17	Build the seller incoming-orders view with tests for correct order-to-seller mapping.	IT25101657 (Order owner)	7
T-03.5	PBI-18	Build order-status update logic (pending/shipped/delivered) with state-transition tests.	IT25101657 (Order owner)	7
T-03.6	PBI-19	Build the buyer order-history view with tests for correct record scoping.	IT25101657 (Order owner)	7
T-03.7	PBI-22	Build listing moderation (flag/remove) with permission and audit tests.	IT25101678 (Admin owner)	7
T-03.8	PBI-23	Build the platform-wide reporting view with tests for data accuracy across users, listings, and orders.	[T25101678 (Admin owner)	7
T-03.9	All Sprint 3 PBIs	Execute order-fulfilment, moderation, regression, and acceptance-evidence checks.	IT25101690 (Developer)	6

4th Sprint

Sprint Goal: Complete remaining profile, listing, sorting, cart, tracking, and complaint-handling features; demonstrate release readiness.

Dependency order: profile and listing enhancements refine Sprint 1–2 foundations; cart sorting and quantity adjustment refine the Sprint 2 checkout flow; order tracking and complaint handling close the loop opened in Sprint 3.

Selected User Stories

Sprint ID	PBI	User Story	Priority
SP4-01	PBI-04	As a registered user, I want to edit my profile details so that my account information stays accurate.	Low
SP4-02	PBI-08	As a seller, I want to delete a listing I no longer sell so that my catalogue stays current.	Low
SP4-03	PBI-11	As a buyer, I want to sort products by price, newest, or rating so that I can compare choices easily.	Medium
SP4-04	PBI-14	As a buyer, I want to adjust the quantity of an item in my cart so that I order the correct amount.	Medium
SP4-05	PBI-20	As a buyer, I want to check the current status of an active order so that I know when to expect delivery.	Low
SP4-06	PBI-24	As an administrator, I want to review escalated complaints so that disputes between buyers and sellers are resolved fairly.	Low

Sprint Backlog — Task Assignments and Ownership

Task ID	Linked PBI	Task	Owner	Hrs
T-04.1	All selected PBIs	Set final backlog order and release criteria, review evidence, and accept or reject the increment.	IT25101719 (Product Owner)	4
T-04.2	PBI-04	Build profile-editing functionality with validation and unit tests.	IT25101719 (Account owner)	7

T-04.3	All selected PBIs	Facilitate release-readiness, Scrum events, final risk review, retrospective, and impediment removal.	IT25101711 (Scrum Master)	4
T-04.4	PBI-08	Build listing-deletion (archive) logic with confirmation and tests.	IT25101711 (Listing owner)	7
T-04.5	PBI-11	Build sort-by (price/newest/rating) functionality with tests for correct ordering.	IT25101690 (Search owner)	7
T-04.6	PBI-14	Build cart quantity-adjustment logic with stock-limit tests.	IT25101736 (Cart owner)	7
T-04.7	PBI-20	Build the current order-status tracking view with tests for real-time accuracy.	IT25101657 (Order owner)	7
T-04.8	PBI-24	Build complaint review and resolution workflow with permission and audit tests.	IT25101678 (Admin owner)	7
T-04.9	All Sprint 4 PBIs	Execute recovery, regression, security, and release-acceptance checks.	IT25101678 (Admin owner)	6

Member Ownership Summary

Member	Role	Owned Task IDs	Total Hrs
[Member 1 Name]	Product Owner	T-01.1, T-01.2, T-02.1, T-03.1, T-03.2, T-04.1, T-04.2	33
IT25101711	Scrum Master	T-01.3, T-01.4, T-02.2, T-02.3, T-02.4, T-03.3, T-04.3, T-04.4	40
IT25101690	Developer	T-01.6, T-01.7, T-02.5, T-03.9, T-04.5	34
[Member 4 Name]	Developer	T-01.8, T-02.6, T-02.7, T-02.8, T-04.6	33
[Member 5 Name]	Developer	T-02.9, T-03.4, T-03.5, T-03.6, T-04.7	34
[Member 6 Name]	Developer	T-01.5, T-03.7, T-03.8, T-04.8, T-04.9	33

Estimated hours are a planning aid. Sprint progress should be reviewed during Daily Scrum and adjusted transparently when new dependencies or risks appear.